Đinh Trung Quốc Anh

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RESEARCH INTERESTS

Star Formation History of Massive Galaxies, Dusty Star-forming Galaxies and Galaxy Morphology at High Redshifts.

EDUCATION

International University, Vietnam National University - Ho Chi Minh City Vietnam
B.Eng in Space Engineering Sep, 2022 - Oct, 2026
All academic activities are in English
Cumulative GPA: 92.5/100 (3.87/4.00)
Thesis Title: Star Formation History of Massive Galaxies at z $>$ 3, Using Non-parametric & Parametric Models
Advisor: Dr. Abdurro'uf (Indiana University)

RESEARCH EXPERIENCE

Independent Research Project: Spatially Resolved Star Formation History of Dusty Merging Galaxies at z $>$ 2 from JWST+ALMA
Advisor: Dr. Abdurro'uf (Indiana University) Aug, 2026 - Present

- **Goal**: Understanding the impact of "wet" mergers on the star formation history (SFH) at Cosmic Noon.
- **Method**: Used piXedfit, Bagpipes and Dense Basis for fitting spatially resolved SED & Analyzed JWST/NIRSpec + ALMA data to investigate the interplay of dust & gas in these extreme systems.

2026 Summer Internship: Morphological Properties of Galaxies in a z = 4.53 Extended Galaxy Structure
Advisors: Prof. Arianna S. Long & PhD Student Nguyễn Bình (University of Washington) Jul, 2026 - Present

- **Goal**: Quantifying the morphology of over 200 galaxies in an extended galaxy structure at z = 4.53.
- **Method**: Used statmorph to derive non-parametric and parametric morphological parameters in F277W + F444W filters & analyze the morphological trends in the extended galaxy structure.

Bachelor Thesis: Star Formation History of Massive Galaxies at z $>$ 3, Using Non-parametric & Parametric Models
Advisor: Dr. Abdurro'uf (Indiana University) Jan, 2026 - Jun, 2026

- **Goal**: Constraining the star formation history (SFH) of a sample of 80 massive galaxies at z $>$ 3 and comparing the effects of non-parametric/parametric SFH models.
- **Method**: Used Bagpipes for SED fitting, with 3 SFH models (Delayed Exponential, Double-Power Law, Non-parametric using Continuity priors); Derived the formation, quenching timescale from non-parametric SFHs.

Final-year Research Project: Measuring Brightness & Fitting the Spectral Energy Distribution of a Lensed Galaxy at z = 3.29
Advisor: Dr. Phạm Tuấn Anh (Vietnam National Space Center) Sep, 2025 - Jan, 2026

- **Goal**: Measuring the flux and SED-fit properties of a gravitationally lensed, high-redshift, Lyman-alpha emitting galaxy P198.
- **Method**: Used SAOImageDS9 and Photutils to measure photometry of HST and JWST images; Applied SED fitting using CIGALE; Examined 2 star-formation history models: sfhdelayed and sfh2exp.

Third-year Research Project: Exploring the Interactions between Martian Crustal Magnetic Field and Solar Winds with MAVEN/SWEA
Advisor: Dr. Jaeheung Park (Korea Astronomy and Space Science Institute) Dec, 2024 - Jun, 2025

- **Goal**: Visualizing the altitude-resolved martian magnetic field's 2D topology and its correlation with the inward/outward flows of solar-wind electrons/photoelectrons.

- **Method**: Calculated the shape parameter for topology classification with MAVEN/SWEA data & Identified the direction of electron populations with pitch angle & magnetic field's radial component.

2024 Summer Internship: Statistical Analysis of Ionospheric Irregularity Distribution Using Swarm Data
Advisors: Dr. Wookyoung Lee (Korea Astronomy and Space Science Institute) & Dr. Hyosub Kil (Johns Hopkins University) Jul, 2024 - Sep, 2024

- **Goal**: Identifying the scintillations at the high-latitude regions during summer of each hemisphere.
- **Method**: Calculated the S parameter - an indicator for ionospheric irregularity; Derived the daytime and nighttime distribution of S parameter and plasma density in a year using Swarm LP data.

2023 Summer Internship: Front-end Testing and Deployment of BURSTT
Advisor: Dr. Chao-Te Li (The Academia Sinica Institute of Astronomy and Astrophysics) Jul, 2023 - Sep, 2023

- **Goal**: Understanding the front-end design for BURSTT radio telescope to detect Fast Radio Bursts.
- **Method**: Assembled the Low-Noise Amplifier & Deployed the first 2x16 antenna array at the Fushan site with BURSTT team; Modelled a simple antenna model with Ansys-HFSS.

OBSERVING EXPERIENCE

Quantifying the Rotation of Saturn by Doppler Spectroscopy
Advisor: Dr. Phan Thanh Hiền (University of Science & Technology of Hanoi) Quy Nhon Observatory, Oct 2026

- Observed the tilted absorption lines of Saturn's spectrum with a 60-cm telescope at Quy Nhon Observatory.
- Reduced the spectrocopic data with Python and Demetra software; Measured the rotational velocity of Saturn by the Doppler shift of the H-alpha line.

Multicolor Observations of Globular Cluster NGC 2419 Using The 80-cm Twin Telescope (Teide Observatory, Spain)
Advisor: Prof. Phan Bảo Ngọc (International University, VNU-HCMC) Remote Observing, Dec 2024

- Submitted an observing proposal for NGC 2419 in the SDSS bands u', g', r', i', z'.
- Plotted the color-color, color-magnitude diagrams for NGC 2419.

PUBLICATIONS

1. *A city at the edge of time: Spectroscopic confirmation of a z = 4.53 extended galaxy structure.*
N. Bình, A.S. Long, R. Oh, **D.T.Q. Anh,** et al. 2026, in-prep
2. *Spatially resolved star formation history of dusty merging galaxies at z > 2 from JWST+ALMA.*
D.T.Q. Anh, Abdurro'uf. 2026, in-prep

HONORS & AWARDS

Gold Medal for GPA achievement, International University 2026
Full Scholarship, International University 2022 - 2026
Development Fund's Scholarship, Vietnam National University - Ho Chi Minh City 2022 - 2026
Dean's List, International University 2023, 2025
Silver Honor, International Astronomy and Astrophysics Competition 2025
Valedictorian of Entrance Exam, International University 2022
Certificate of Merit for College/University Valedictiorians, Ho Chi Minh City People's Committee 2022

SKILLS

- **Programming:** PYTHON, C, MATLAB, SWIFT, SQL, HTML, CSS
- **SED-Fitting:** Integrated & Spatially Resolved SED fitting using Bagpipes, CIGALE, piXedfit, Dense-basis
- **Photometry Tools:** APT, Siril, SAO ImageDS9, Photutils, SEP
- **Galaxy Morphology Codes:** statmorph
- **Machine Learning Algorithms:** Random Forest, K-Means, DBScan, CNN
- **Antenna/Radio Simulations:** Ansys-HFSS, GNU-Radio

PARTICIPATED ASTRONOMICAL SCHOOLS

48th International School for Young Astronomers Sep 28 - Oct 16, 2026
Organized by The International Astronomical Union (IAU) & ICISE Gia Lai, Vietnam

SOKENDAI NAOJ Asian Winter School Feb, 2025 & Feb, 2026
Organized by The Graduate Institute for Advanced Studies - SOKENDAI Online

3rd TNU/SAGI Summer School for Observational Astrophysics Jul 27 - Aug 10, 2025
Organized by Simons Astrophysics Group at ICISE (SAGI) & Tây Nguyên University (TNU) Đăk Lăk, Vietnam

SERVICE

Member of Vietnam Amateur Astronomy Network (VAAN)	Aug, 2025 - Present
Class Monitor of K2022, Space Engineering	2022 - 2026
Member of Space Engineering's Admissions Counseling Team	2023 - 2026
Content Creator for 'Preparation for Entrance Exam' Program, Thanh Niên's Newspaper	2023, 2024, 2026
Head of Space Engineering's Astronomy Club (SEAC)	2023 - 2025
Member of Space Engineering's English Speaking Club (SEEC)	2023 - 2025
Co-organizer of Mid-Autumn Moon Gazing Event for World Space Week 2025	Oct, 2025
Calculus Tutor for Space Engineering Students	Oct, 2025 - Dec, 2025
Leader of IDEA Team, IU Guitar Club	Jul, 2023 - Jul, 2024

TALKS & PRESENTATIONS

Non-parametric/parametric star formation histories of massive galaxies at $z > 3$ Aug, 2026
For University of Washington & University of Manchester Research Group Meeting Online

Introduction to Astro-Imaging Nov, 2025
For Seminar Hosted by Space Engineering's Astronomy Club Ho Chi Minh City, Vietnam

REFERENCES

Prof. Arianna S. Long (*Department of Astronomy, University of Washington*)
E-mail: aslong@uw.edu

Dr. Abdurro'uf (*Department of Astronomy, Indiana University*)
E-mail: fnuabdur@iu.edu

Dr. Phạm Tuấn Anh (*Department of Astrophysics, Vietnam National Space Center*)
E-mail: ptanh@vnsc.org.vn

Dr. Phan Hiền Vũ (*Department of Physics, International University*)
E-mail: phvu@hcmiu.edu.vn